



VISUAL ANALYTICS: UNCERTAINTY AND BIAS LAB

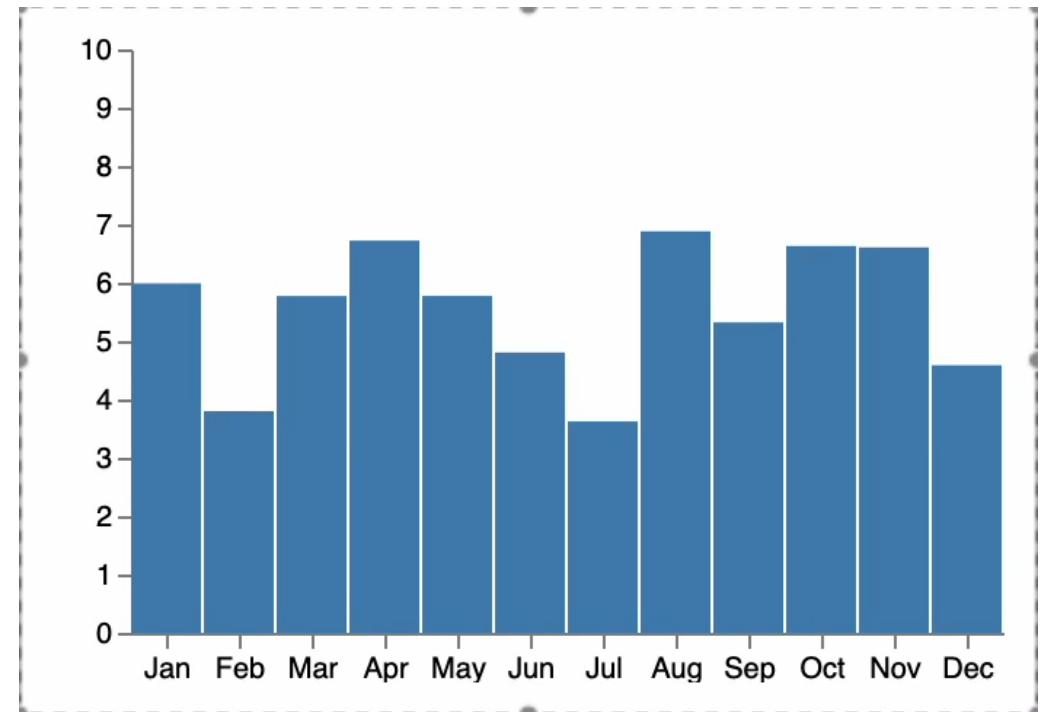
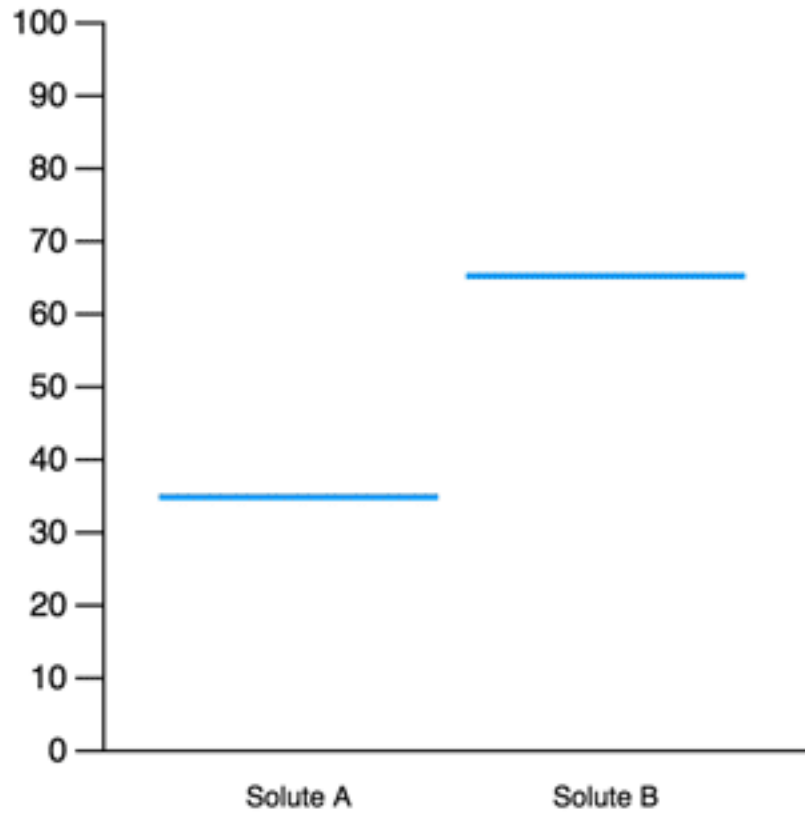
AB MOSCA (THEY/THEM)



TODAY'S PLAN

- Refresh: HOPS
 - HOPS Algorithm
- Refresh: Bias in visual analyses
- Visualizing bias in dataset exploration

HYPOTHETICAL OUTCOME PLOTS



<https://medium.com/hci-design-at-uw/hypothetical-outcomes-plots-experiencing-the-uncertain-b9ea60d7c740>

<https://vega.github.io/vega/examples/hypothetical-outcome-plots/>

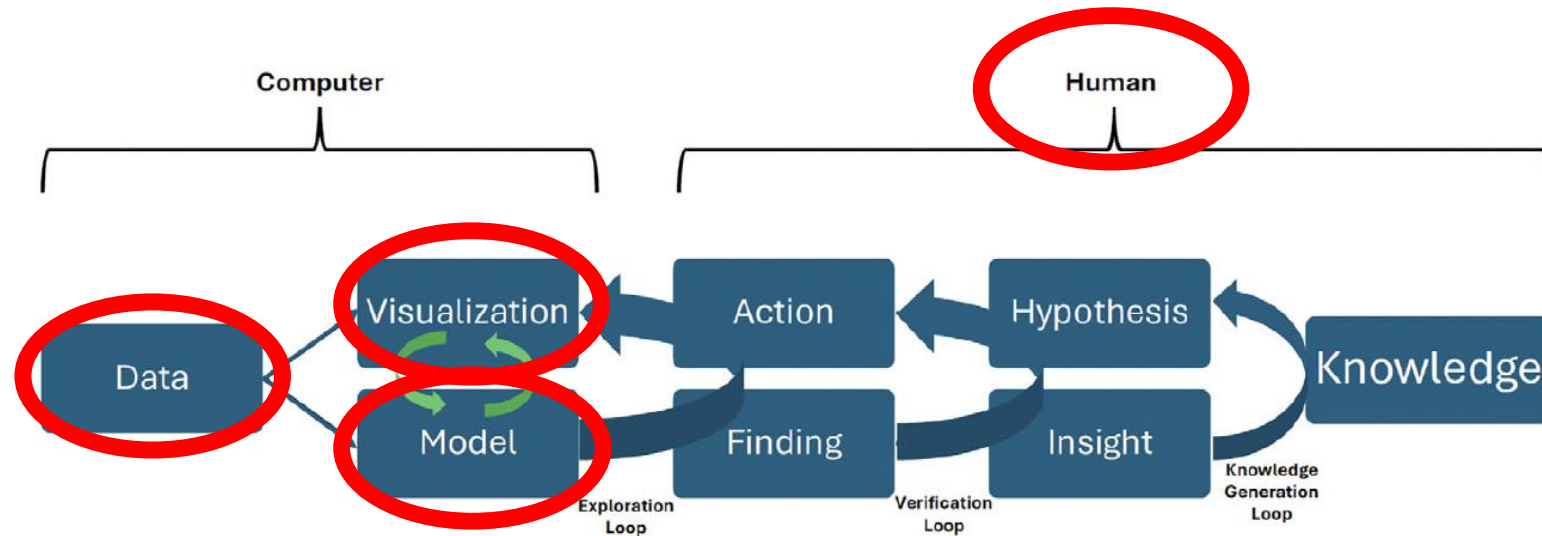
HYPOTHETICAL OUTCOME PLOTS

HOP Algorithm

1. Draw a sample of hypothetical outcomes (draws) from a distribution
 - Bootstrapping (resample original data with replacement)
 - Infer a distribution model
2. For each, make a plot that becomes one frame in an animation
 - Function to plot data with consistent axes, encoding, etc.
 - Function to update (animate) between hypothetical outcomes

BIAS IN VISUAL ANALYTICS

- What do you remember? Write down 3 specific things.



BIAS IN VISUAL ANALYTICS

Real-Time interventions

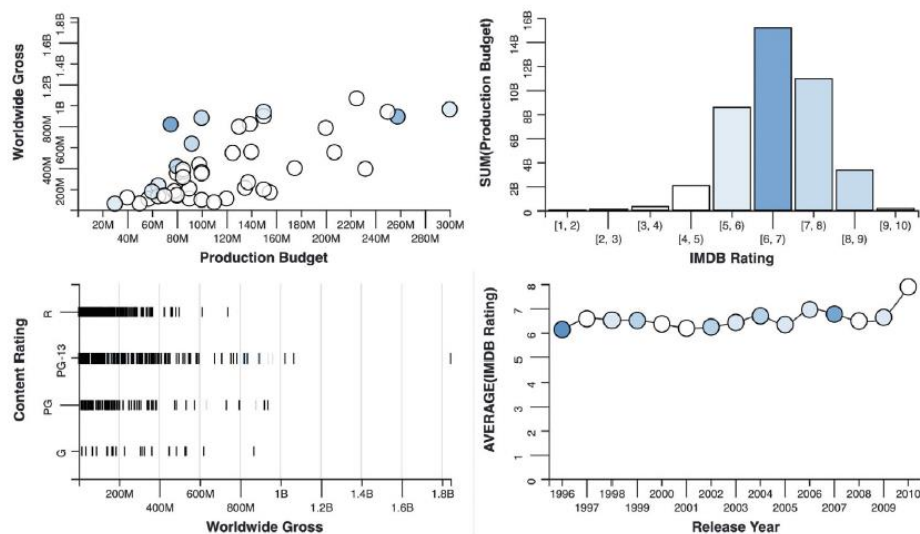


Fig. 7.5 Lumos captures mouseover interactions on visualization elements (e.g., points, bars) and leaves traces in-situ by coloring (fill) them in shades of blue in proportion to the relative frequency of interactions

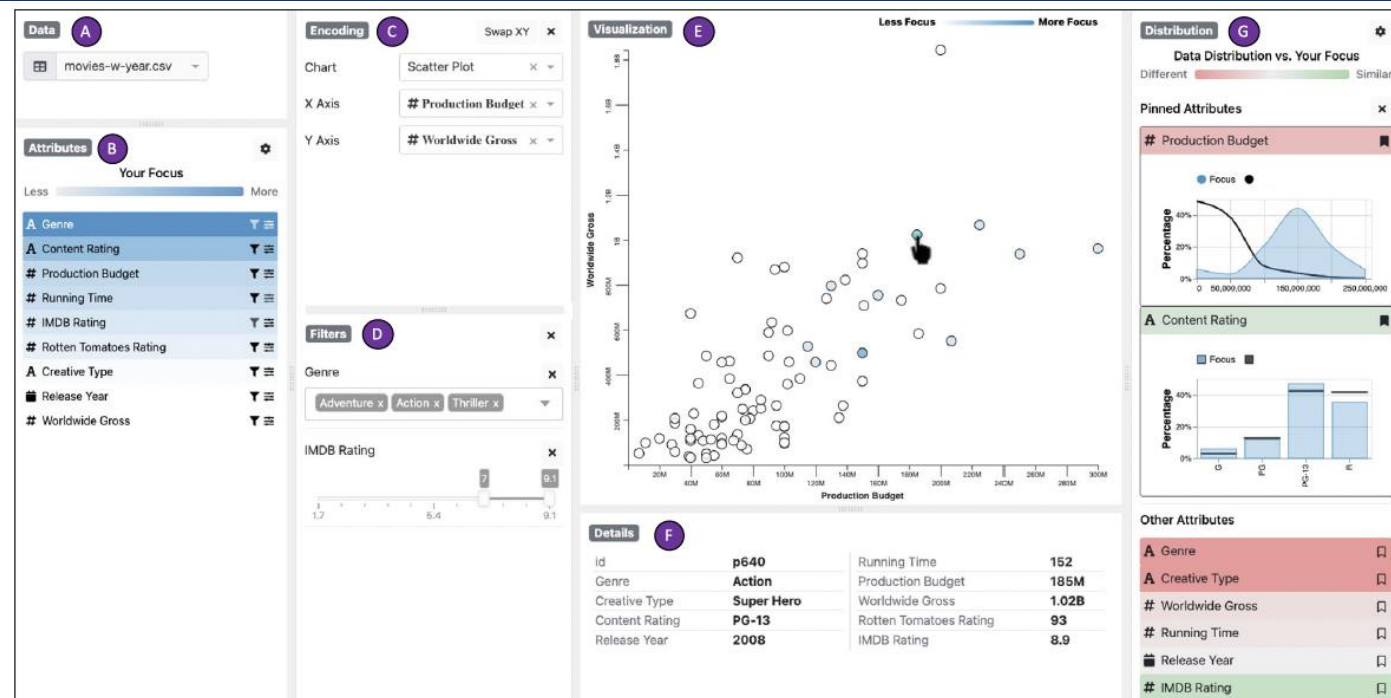


Fig.7.3 The Lumos user interface includes traditional visual data analysis functions ((A) **Data Panel**, (B) **Attributes Panel**, (C) **Encoding Panel**, (D) **Filter Panel**), and shows analytic behavior as in-situ and ex-situ interaction traces in the (B) **Attributes Panel**, (E) **Visualization Canvas**, (F) **Details View**, and (G) **Distribution Panel** as the relation between a target distribution (e.g., the underlying data) and the user's analytic behavior

VISUALIZING DATASET EXPLORATION

- In the starter code for today's lab, you will find a coordinated multiple view (CMV) visualization of the palmer's penguins dataset
- On the left scatterplot, there is a brush that allows the user to select a subset of points
- Selected points are highlighted in red on both plots
- Modify the starter code such that:
 1. Interaction traces are shown. In other words, how can you keep track of and visualize which points have been selected previously?
 2. Add a bar plot that shows number of each penguin species in a selection